

STUDENT-CONSTRUCTED CENTRIFUGE

Lessons Learned

This Lessons Learned document focuses on a classroom incident in which a small purpose-built ultracentrifuge failed. The shrapnel from the device injured a student's hand, requiring stitches. It demonstrates the importance of reviewing the safety of the interim steps involved in a project-based course, not just the safety of the final product, as well as the importance of properly containing hazardous equipment.

What Happened?

Several students were working in a course that involved design and construction of centrifuges. This course served as a "pre-capstone" course, helping students develop the skills they will need for their actual capstone (in which senior students choose or are assigned a significant independent project, which requires them to use all the knowledge and skills they developed in their prior courses). In this pre-capstone course, students began by building a centrifuge following very specific instructions and were gradually given more autonomy throughout the course, culminating in them working in small teams over the last two weeks of the course to design and implement a significant modification to their centrifuge. These designs are reviewed by the course instructor and course assistants (CAs). At the time of this incident, the safety portion of this review focused on the safety of the final product, but did not fully consider the safety of the construction and testing process.

The students involved in this event chose to design and build an ultracentrifuge. Ultracentrifuges are designed to spin at very high speeds; this allows them to generate greater centrifugation forces, but they are therefore more hazardous. [When commercial ultracentrifuges fail](#), they can cause significant destruction to the area around them and can cause significant injury. The team was aware of the hazards associated with ultracentrifuges, and were building one at a smaller scale than those found in labs. Its maximum rotation speed was designed to be 20,000 rpm, which is low relative to commercial ultracentrifuges (which often operate between 60,000 and 200,000 rpm). The rotor of the ultracentrifuge, the part that spins and is thus subjected to significant g forces, was designed to be relatively slim and light-weight to allow it to achieve the necessary speed using a commercially available low-cost hobby motor.

In the final week of the course, two student team members were in the teaching lab, working on the ultracentrifuge. Part of the procedure in the course is to test components and subsystems, and on this day they were testing the rotor and motor. This entailed connecting the rotor to the motor and spinning it without the unit's protective housing (which was part of the design but was not yet built). The course instructor told the team to cover the setup with something, such as a clear plastic bin, to enable observation of the test while creating a safety barrier. When the students tested the setup, however, it was not covered. The course instructor was not in the room, and the CA present was assisting another team. One of the team members, the student operator, turned on the setup and began ramping up the rotation speed. They felt

uncertain about going too fast, but were urged by other students to increase the speed. They ramped the speed toward 20,000 rpm; during the ramping of the speed, the rotor was torn apart by the shear forces it was subjected to, flinging shrapnel up to 6.5 feet (2 meters) away, some of which hit the operator's hand (which was directly adjacent to the rotor, at this interim stage in the design).

The student operator initially did not feel any pain, and turned the apparatus off. A CA heard the noise and came to investigate. The student operator noticed several cuts on their hand, most of which were surface level but one of which seemed to be deeper. They went to the bathroom to wash their hands, and concluded that the deeper cut was indeed more serious. After leaving the bathroom, the student operator, the other team member, and the CA met with the course instructor. The instructor drove the student operator to the Stanford Hospital Emergency Department, where the student operator received 7 stitches.

Why Did This Happen?

The immediate cause of the incident was a structural failure of the centrifuge's rotor due to the shear forces it was subjected to, in particular failing in a way that it fragmented and flung the fragments out, causing injury. The slim design of the rotor (which allowed it to achieve the necessary speed) and its brittle material of construction meant it was not able to withstand the significant forces created by spinning at approximately 15,000 rpm, below the original design's requirement.

Contributing factors to the immediate cause included the setup not being covered and the speed being ramped too quickly. The device should have been tested more thoroughly at lower speeds before being ramped to its maximum rotational speed of 20,000 rpm. While this may not have prevented the incident, given the abrupt nature of the failure mode, but may have provided opportunities to realize and fully appreciate the risks involved.

The root cause of the incident was insufficient risk assessment of the testing process. While the safety of the "final product" was assessed by both the students and the teaching team as part of course assignments, these assessments did not consider the risks associated with the testing process (i.e., that structural damage or failure could cause injuries to those conducting the test). Furthermore, while the risk of failure due to shear forces was identified and considered, they were not fully assessed or modeled, as this would be beyond the scope of the class (the modeling of 3D printed plastic parts is difficult even when within scope). Thus, while both the students and the teaching staff were aware of the hazards associated with constructing an ultracentrifuge in a general sense, there was no process within the course's structure to assess and control the intermediate hazards associated with construction or testing hazards. Additionally, the course did not have policies regarding what level or types of hazard requires CA and/or instructor supervision.

Another factor of note was the "peer pressure" the operator felt during the incident. While none of their classmates intentionally encouraged the operator to act unsafely (i.e., no one explicitly told them to ignore safety considerations), some of their comments and the general feeling of group excitement led to the operator feeling some pressure to continue the test and to push the limits of their setup. While this incident may have occurred regardless, this pressure did increase the likelihood of risky actions.

What Went Right?

A few factors prevented this from being a more serious incident. Perhaps most importantly, all shrapnel from the rotor's failure traveled in a straight horizontal path. Had any traveled at an upward angle, or ricocheted unpredictably, it could have struck students in the eyes, neck, and/or face. Another mitigating factor was

that everyone involved remained calm and efficient, leading to the situation being addressed in a prompt and effective manner.

In the aftermath of the incident, the students and instructors worked closely with EH&S to analyze the incident and to determine ways to prevent similar incidents. This will lead to improved safety not just in this course, but in similar courses across campus.

How to Prevent Incidents Like This

EH&S has a number of recommendations and resources for preventing both incidents like this and incidents in general. The instructors for the course also devised and implemented improvements in their practices and operating procedures.

1. Instructors should incorporate safety assessment of experimental procedures as an integral part of lab courses, and should review students' assessments closely. This is especially important for advanced courses where students develop their own procedures.
 - a. In lab courses involving prototyping of devices and component testing, safety assessment of testing procedures should be included.
2. Instructors for advanced courses where students develop their own procedures should establish clear rules regarding procedures and conditions that require written safety plans, instructor supervision, EH&S review, and other enhanced safety measures.
 - a. Instructors can review EH&S's [Laboratory Risk Assessment Tool](#) for help determining what hazards to account for and how to categorize risk levels. EH&S is available to assist in development of course guidelines.
 - b. Procedures for testing equipment/components should include clear steps for ramping tests in a controlled manner.
3. Instructors should provide and require the use of safety barriers for procedures that could result in flying debris.
4. Instructors should require the use of eye protection (e.g., safety glasses) for all students in a lab whenever any one student/group of students is performing a procedure that could result in flying debris.
5. Instructors and other teaching staff should ensure that students are comfortable with the tasks they are performing and empower students to stop work if they feel or notice something unsafe. Some ways instructors can accomplish this is by encouraging questions about safety and by regularly reminding students that they are authorized to stop work if they see something that concerns them.